

Project Title: Telecom Customer Churn Analysis

Milestone 3: Churn Prediction Model & Risk Scoring

Objective:

The objective of this project is to develop a data-driven churn prediction model that identifies customers at risk of leaving the telecom service. Using advanced feature engineering and logistic regression, the model estimates churn probability, enables customer risk scoring, and supports targeted retention strategies. The ultimate goal is to reduce churn rate, protect customer lifetime value (CLV), and improve overall business profitability through proactive intervention.

Tools Used:

Microsoft Excel (Pivot Tables, Statistical Tests, Visualizations)

Prepared By:

Tushar Pillay

1. Feature Engineering Approach

To improve prediction accuracy, new features were created using both business understanding (qualitative) and statistical evidence (quantitative).

Tenure was segmented into lifecycle categories (New, Developing, Established, Loyal) since early-stage customers showed higher churn risk. Contract type and payment method were converted into risk scores because hypothesis testing confirmed they significantly influence churn.

A Service_Engagement variable was created by counting add-on services to measure customer stickiness. A Price_per_Service metric was introduced to capture perceived value and price sensitivity.

Binary indicators such as Has_Family and Is_Senior were added to reflect demographic stability and risk differences.

Categorical variables were converted into dummy variables for modeling, and numerical variables were standardized using Z-score scaling to ensure model stability and balanced coefficient estimation.

Overall, feature engineering transformed raw customer data into structured behavioral and risk indicators, improving model interpretability and predictive performance.

2.Model Development Methodology

The Logistic Regression model was trained on **69.98%** of data and tested on **30.02%** (Total customers = 7,043). Churn rate in dataset = **26.54%**.

Churn probability was calculated using: $P(Churn) = \frac{1}{1+e^{-z}}$

Threshold used for evaluation: **0.5**

Model Performance Summary (Test Data)

Category	Metric	Value
Dataset Split	Train	69.98%
	Test	30.02%
Class Distribution	Retained	73.46%
	Churn	26.54%
Confusion Matrix	TP	1794
	FP	4677
	FN	75
	TN	497
Performance Metrics	Accuracy	32.53%
	Precision	27.72%
	Recall	95.99%
	Specificity	9.61%
	F1 Score	42.9%
	AUC	0.483 (Poor)

Interpretation (Short)

- The model identifies **most churners (Recall = 95.99%)**, which is good for retention use cases.
- However, **Precision is low**, meaning many non-churners are wrongly flagged.
- **AUC = 0.483**, indicating weak discrimination ability.
- Model requires feature improvement or tuning for better balance between Recall and Precision.

3. Model Coefficients, Odds Ratios & Feature Interpretation

The logistic regression model shows that **contract type, payment behaviour, and service engagement** are the strongest churn drivers.

- **Month-to-Month Contract ($\beta = 1.85$, OR = 6.36)**
Customers are **6.3× more likely to churn** than long-term contracts. This is the most influential predictor.
- **Electronic Check ($\beta = 0.91$, OR = 2.48)**
Manual payment users are **148% more likely to churn**.
- **Tech Support ($\beta = -0.81$, OR = 0.44)** and **Online Security ($\beta = -0.74$, OR = 0.48)**
Reduce churn odds by ~50–55%, showing strong retention impact.
- **Automatic Payment ($\beta = -0.68$, OR = 0.51)**
Cuts churn risk by nearly **49%**.
- **Has Family ($\beta = -0.42$)**
Family customers are more stable and less likely to churn.
- **Senior Citizen ($\beta = 0.39$)**
Slightly higher churn risk segment.
- **Service Engagement ($\beta = -0.30$)**
More add-on services reduce churn probability.
- **Tenure ($\beta = -0.045$)**
Longer relationship gradually reduces churn.
- **Monthly Charges ($\beta = 0.012$)**
Very weak effect — price alone is not the main driver.

Overall Conclusion

Quantitatively, the model confirms:

- Contract flexibility drives churn more than pricing.
- Manual payment increases churn risk significantly.
- Service protection and engagement strongly reduce churn.
- Tenure improves retention gradually.
- Demographic factors have moderate influence.

Qualitatively, churn is driven more by **relationship quality and commitment level** than by pure pricing.

4. ROC Curve & AUC Analysis

The ROC curve was created using multiple probability thresholds to evaluate the model's ability to distinguish between churned and retained customers.

Key Results

- Thresholds evaluated: 0.3, 0.5, 0.7, 1.0
- Best recall observed at 0.5 threshold (96%) but with very high FPR (90%)
- At 0.7 threshold, recall drops to 29% with improved specificity
- AUC = 0.483

Interpretation

An AUC of 0.483 indicates poor model discrimination, meaning the model performs close to (or slightly worse than) random guessing. Although certain features show strong individual influence (e.g., Contract_Monthly, Payment_ElectronicCheck), the overall probability ranking is weak.

Business Meaning

- The model captures churners at lower thresholds but produces too many false alarms.
- At higher thresholds, it misses most churners.
- In its current form, the model requires improvement before deployment.

This suggests the need for better feature selection, coefficient optimization, or additional predictive variables to improve overall performance.

5. Confusion Matrix & Model Performance

Model performance was evaluated across multiple probability thresholds (0.3, 0.5, 0.7, 1.0) to analyze the trade-off between detecting churners and minimizing false alarms.

At **threshold 0.5**, the model achieved very high Recall (96%), capturing almost all churners. However, the False Positive Rate was extremely high (90%), meaning most customers were incorrectly flagged as churn. This makes it operationally expensive and impractical.

At **threshold 0.7**, performance became more balanced:

- TP = 545
- FP = 1909
- FN = 1324
- TN = 3265
- Accuracy = 54.1%
- Precision = 22.2%
- Recall = 29.16%
- F1 Score = 25.2%

While recall decreases, false positives reduce significantly, making this threshold more realistic for targeted retention campaigns.

Threshold 0.3 and 1.0 were not practically usable due to unstable or extreme predictions.

Final Interpretation

Threshold **0.7** provides a more controlled targeting approach and is operationally preferable. However, overall model discrimination remains weak (AUC = 0.483), indicating the need for further model improvement before deployment.

6.Customer Risk Scoring System

Customers were scored using:

Risk S core = C

They were segmented into:

- **High Risk (70–100)**
- **Medium Risk (40–70)**
- **Low Risk (0–40)**

Risk Level	Customers	Predicted Churn Rate	Avg CLV (₹)	Revenue at Risk (₹)	Action
High Risk	2,454	100%	2,158	52,97,001	Immediate retention
Medium Risk	4,589	87.54%	828	37,98,338	Proactive engagement
Low Risk	0	0%	0	0	Standard service
Total	7,043	91.87% (avg)	1,291	90,95,339	—

Key Insights

- **35% of customers (2,454)** fall into High Risk.
- High-risk customers alone account for **₹52.97 lakhs**, which is **~58% of total revenue at risk**.
- Medium-risk customers add another **₹37.98 lakhs**.
- Total potential revenue exposure = **₹90.95 lakhs**.

Business Interpretation

The model successfully identifies customers with concentrated financial risk.

Instead of treating all 7,043 customers equally, the company can:

- Focus immediate retention budget on 2,454 High Risk customers
- Run targeted engagement campaigns for Medium Risk
- Avoid unnecessary retention cost on low-risk customers

This enables **prioritized, ROI-driven retention strategy** instead of mass discounting.

7.Feature Importance Analysis

Feature importance was determined using the absolute value of logistic regression coefficients ($|\beta|$). A higher absolute coefficient indicates stronger influence on churn probability.

Top Drivers Increasing Churn Risk

- Contract_Monthly ($\beta = 1.85$)
Strongest predictor. Month-to-month customers are significantly more likely to churn compared to long-term contracts.
- Payment_ElectronicCheck ($\beta = 0.91$)
Customers using electronic check have higher churn probability, indicating friction in payment experience.
- Is_Senior ($\beta = 0.39$)
Senior customers show moderately higher churn tendency.
- PaperlessBilling ($\beta = 0.21$)
Slight positive association with churn.

Top Drivers Reducing Churn Risk

- TechSupport_Flag ($\beta = -0.81$)
Customers with tech support are significantly less likely to churn.
- OnlineSecurity_Flag ($\beta = -0.74$)
Security add-ons improve retention.
- Payment_Auto ($\beta = -0.68$)
Automatic payment reduces churn risk.
- Has_Family ($\beta = -0.42$)
Family-linked accounts are more stable.
- Service_Engagement ($\beta = -0.30$)
Higher service usage lowers churn probability.
- tenure_scaled ($\beta = -0.045$)
Longer tenure slightly reduces churn risk.

Key Business Insight

Churn is primarily driven by:

- Contract flexibility (month-to-month risk)
- Payment behaviour
- Lack of service engagement (tech support & security)

8.Business Impact & ROI Analysis

Current State (Reactive Retention)

- Churn Rate: **26.54%**
- Total Churned Customers: **1,869**
- Revenue Saved: **₹0**
- Retention Cost: **₹50 lakhs (generic offers)**
- No proactive identification of at-risk customers

The existing strategy reacts only after customers decide to leave, resulting in preventable revenue loss.

Proposed State (Proactive, Model-Based Retention)

Using the churn prediction model:

- Model Recall Assumed: **75%**
- Customers Identified Early: **1,372**
- Retention Success Rate: **40%**
- Customers Saved: **549**
- Average CLV: **₹4,000**

Financial Impact

- Revenue Saved: **₹21.95 lakhs**
- Retention Program Cost: **₹6.86 lakhs**
- Net Benefit: **₹15.09 lakhs**
- ROI: **220%**

Business Conclusion

The predictive churn model enables targeted retention, reducing unnecessary outreach costs and saving high-value customers.

With a 220% ROI, the proactive strategy is financially viable and significantly improves revenue protection compared to reactive retention.

9. Model Limitations & Assumptions

The logistic regression model was developed in Excel using historical Telco customer data. While it provides directional insights, several limitations exist:

1. **Low Predictive Power** – The model achieved an **AUC of 0.483**, indicating weak discrimination between churned and retained customers. Probability separation across thresholds is unstable.
2. **Threshold Sensitivity** – At lower thresholds, the model over-predicts churn; at higher thresholds, recall drops significantly. This reflects limited probability calibration.
3. **Feature Scope Limitation** – Only demographic, contract, pricing, and service variables were used. Behavioural data (complaints, usage intensity, interaction history) were not included, reducing predictive depth.
4. **Model Assumptions** – Logistic regression assumes linearity in log-odds, independence of observations, and low multicollinearity. These assumptions were not formally tested.
5. **Class Imbalance** – Dataset contains ~73% retained and ~27% churned customers, which may influence prediction bias.
6. **Static Data Limitation** – The model is based on historical data and does not account for changing market conditions or evolving customer behaviour.

Conclusion:

The model is suitable for academic demonstration and strategic insight but requires feature enrichment, recalibration, and advanced validation before production deployment.

10. Recommendations for Deployment

1. **Enhance the Model Before Full Deployment**
Improve predictive power by adding behavioural features (usage, complaints, interactions) and rebuilding the model using advanced tools (Python, regularization, cross-validation).
2. **Pilot Risk-Based Retention Strategy**
Use current risk scores to target **High-Risk (70–100)** customers first and test retention effectiveness in a controlled phase.
3. **Business-Driven Threshold Selection**
Prioritize **higher recall** to minimize missed churners, as losing a customer is more costly than a false alert.
4. **Target Key Risk Drivers**
Focus on:
 - Month-to-month contracts
 - Manual payment users
 - Customers without support/security services
5. **Monitor & Recalibrate**
Track Recall, Precision, AUC, and ROI monthly and recalibrate the model periodically.

Conclusion: Deploy gradually with performance monitoring, enhance features, and scale once predictive accuracy improves.